ESXi

SAN Host

NetApp
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ESXi

適用於ESXi 7.x的NVMe主機組態、ONTAP 含功能不全

支援能力

NVMe over Fibre Channel（NVMe / FC）支援ONTAP 於NVM9.7或更新版本。

功能

- ESXi啟動器主機可透過相同的介面卡連接埠、同時執行NVMe/FC和FCP流量。請參閱 "Hardware Universe" 以取得支援的FC介面卡和控制器清單。請參閱 "NetApp 互通性對照表" 以取得最新的支援組態和版本清單。
- 從VMware支援的版本號為0、99.1、P3開始ONTAP 、ESXi 7.0更新版本3支援NVMe / FC功能。
- 對於ESXi 7.0及更新版本、HPP（高效能外掛程式）是NVMe裝置的預設外掛程式。

已知限制

不支援下列組態：

- RDM對應
- VVols

啟用NVMe/FC

1. 檢查ESXi主機NQN字串、並確認其與ONTAP VMware ESXi陣列上對應子系統的主機NQN字串相符：

```
# esxcli nvme  info get
Host NQN: nqn.2014-08.com.vmware:nvme:nvme-esx

# vserver nvme subsystem host show -vserver vserver_nvme
  Vserver Subsystem             Host NQN
  ------- ------------------- ----------------------------------------
  vserver_nvme ss_vserver_nvme nqn.2014-08.com.vmware:nvme:nvme-esx
```

設定Broadcom / Emulex

1. 請參閱、檢查所需的驅動程式/韌體是否支援該組態 "NetApp 互通性對照表"。
2. 設定lffc驅動程式參數「lffc_enable_FC4_type=3」、以便在「lffc"驅動程式中啟用nvm/FC支援、然後重新啟動主機。

> 從vSphere 7.0更新3開始、「brcmnvmefc"驅動程式已無法使用。因此、「lffc」驅動程式現在包含先前隨「brcmnvmefc」驅動程式提供的NVMe over Fibre Channel（NVMe / FC）功能。

LPe35000系列介面卡預設會設定「lfc_enable_FC4_type=3」參數。您必須針對LPe32000系列和LPe31000系列介面卡手動設定下列命令。

```
# esxcli system module parameters set -m lpfc -p lpfc_enable_fc4_type=3

#esxcli system module parameters list  -m lpfc | grep lpfc_enable_fc4_type
lpfc_enable_fc4_type              int     3      Defines what FC4 types
are supported

#esxcli storage core adapter list
HBA Name  Driver   Link State  UID
Capabilities         Description
--------  -------  ----------  ------------------------------------
-------------------  -----------
vmhba1    lpfc     link-up     fc.200000109b95456f:100000109b95456f
Second Level Lun ID  (0000:86:00.0) Emulex Corporation Emulex LPe36000
Fibre Channel Adapter    FC HBA
vmhba2    lpfc     link-up     fc.200000109b954570:100000109b954570
Second Level Lun ID  (0000:86:00.1) Emulex Corporation Emulex LPe36000
Fibre Channel Adapter    FC HBA
vmhba64   lpfc     link-up     fc.200000109b95456f:100000109b95456f
(0000:86:00.0) Emulex Corporation Emulex LPe36000 Fibre Channel Adapter
NVMe HBA
vmhba65   lpfc     link-up     fc.200000109b954570:100000109b954570
(0000:86:00.1) Emulex Corporation Emulex LPe36000 Fibre Channel Adapter
NVMe HBA
```

設定**Marvell / QLogic**

1. 請參閱、檢查所需的驅動程式/韌體是否支援組態 "NetApp 互通性對照表"。
2. 將「qlnativefc」驅動程式參數「ql2xnvmesupport=1」設定為在「qlnativefc」驅動程式中啟用nvm/FC支援、然後重新啟動主機。

   '# esxcfg-module-s 'ql2xnvmesupport=1' qlnativefc'

   QLE 277x系列介面卡預設會設定「qlnativefc」驅動程式參數。您必須執行下列命令、以手動設定QLE 277x系列介面卡。

   ```
   esxcfg-module -l | grep qlnativefc
   qlnativefc               4    1912
   ```

3. 檢查介面卡是否已啟用NVMe：

```
  #esxcli storage core adapter list
HBA Name  Driver      Link State  UID
Capabilities         Description
--------  ----------  ----------  ------------------------------------
-------------------  -----------
 vmhba3    qlnativefc  link-up     fc.20000024ff1817ae:21000024ff1817ae
Second Level Lun ID  (0000:5e:00.0) QLogic Corp QLE2742 Dual Port 32Gb
Fibre Channel to PCIe Adapter    FC Adapter
vmhba4    qlnativefc  link-up     fc.20000024ff1817af:21000024ff1817af
Second Level Lun ID  (0000:5e:00.1) QLogic Corp QLE2742 Dual Port 32Gb
Fibre Channel to PCIe Adapter FC Adapter
vmhba64   qlnativefc  link-up     fc.20000024ff1817ae:21000024ff1817ae
(0000:5e:00.0) QLogic Corp QLE2742 Dual Port 32Gb Fibre Channel to PCIe
Adapter  NVMe FC Adapter
vmhba65   qlnativefc  link-up     fc.20000024ff1817af:21000024ff1817af
(0000:5e:00.1) QLogic Corp QLE2742 Dual Port 32Gb Fibre Channel to PCIe
Adapter  NVMe FC Adapter
```

驗證NVMe/FC

1. 確認ESXi主機上已列出NVMe / FC介面卡：

```
# esxcli nvme adapter list

Adapter  Adapter Qualified Name           Transport Type  Driver
Associated Devices
-------  -------------------------------  --------------  ----------
------------------
vmhba64  aqn:qlnativefc:21000024ff1817ae  FC              qlnativefc
vmhba65  aqn:qlnativefc:21000024ff1817af  FC              qlnativefc
vmhba66  aqn:lpfc:100000109b579d9c        FC              lpfc
vmhba67  aqn:lpfc:100000109b579d9d        FC              lpfc
```

2. 確認已正確建立NVMe / FC命名空間：

   下列範例中的UUID代表NVMe / FC命名空間裝置。

```
# esxcfg-mpath -b
uuid.5084e29a6bb24fbca5ba076eda8ecd7e : NVMe Fibre Channel Disk
(uuid.5084e29a6bb24fbca5ba076eda8ecd7e)
   vmhba65:C0:T0:L1 LUN:1 state:active fc Adapter: WWNN:
20:00:34:80:0d:6d:72:69 WWPN: 21:00:34:80:0d:6d:72:69  Target: WWNN:
20:17:00:a0:98:df:e3:d1 WWPN: 20:2f:00:a0:98:df:e3:d1
   vmhba65:C0:T1:L1 LUN:1 state:active fc Adapter: WWNN:
20:00:34:80:0d:6d:72:69 WWPN: 21:00:34:80:0d:6d:72:69  Target: WWNN:
20:17:00:a0:98:df:e3:d1 WWPN: 20:1a:00:a0:98:df:e3:d1
   vmhba64:C0:T0:L1 LUN:1 state:active fc Adapter: WWNN:
20:00:34:80:0d:6d:72:68 WWPN: 21:00:34:80:0d:6d:72:68  Target: WWNN:
20:17:00:a0:98:df:e3:d1 WWPN: 20:18:00:a0:98:df:e3:d1
   vmhba64:C0:T1:L1 LUN:1 state:active fc Adapter: WWNN:
20:00:34:80:0d:6d:72:68 WWPN: 21:00:34:80:0d:6d:72:68  Target: WWNN:
20:17:00:a0:98:df:e3:d1 WWPN: 20:19:00:a0:98:df:e3:d1
```

在支援支援支援的支援區塊9.7中ONTAP 、NVMe / FC命名空間的預設區塊大小為4K。此預設大小與ESXi不相容。因此、在建立ESXi命名空間時、您必須將命名空間區塊大小設為512B。您可以使用「vserver NVMe命名空間create」命令來執行此動作。

「Vserver NVMe命名空間create -vserver vvs 1 -path /vol/nslov/namespace1 -size 100g -osttype VMware -block-size 512B」

請參閱 "指令手冊頁ONTAP" 以取得更多詳細資料。

3. 驗證個別NVMe / FC命名空間裝置的個別ANA路徑狀態：

```
esxcli storage hpp path list -d uuid.5084e29a6bb24fbca5ba076eda8ecd7e
fc.200034800d6d7268:210034800d6d7268-
fc.201700a098dfe3d1:201800a098dfe3d1-
uuid.5084e29a6bb24fbca5ba076eda8ecd7e
   Runtime Name: vmhba64:C0:T0:L1
   Device: uuid.5084e29a6bb24fbca5ba076eda8ecd7e
   Device Display Name: NVMe Fibre Channel Disk
(uuid.5084e29a6bb24fbca5ba076eda8ecd7e)
   Path State: active
   Path Config: {TPG_id=0,TPG_state=AO,RTP_id=0,health=UP}

fc.200034800d6d7269:210034800d6d7269-
fc.201700a098dfe3d1:201a00a098dfe3d1-
uuid.5084e29a6bb24fbca5ba076eda8ecd7e
   Runtime Name: vmhba65:C0:T1:L1
   Device: uuid.5084e29a6bb24fbca5ba076eda8ecd7e
   Device Display Name: NVMe Fibre Channel Disk
(uuid.5084e29a6bb24fbca5ba076eda8ecd7e)
   Path State: active
   Path Config: {TPG_id=0,TPG_state=AO,RTP_id=0,health=UP}

fc.200034800d6d7269:210034800d6d7269-
fc.201700a098dfe3d1:202f00a098dfe3d1-
uuid.5084e29a6bb24fbca5ba076eda8ecd7e
   Runtime Name: vmhba65:C0:T0:L1
   Device: uuid.5084e29a6bb24fbca5ba076eda8ecd7e
   Device Display Name: NVMe Fibre Channel Disk
(uuid.5084e29a6bb24fbca5ba076eda8ecd7e)
   Path State: active unoptimized
   Path Config: {TPG_id=0,TPG_state=ANO,RTP_id=0,health=UP}

fc.200034800d6d7268:210034800d6d7268-
fc.201700a098dfe3d1:201900a098dfe3d1-
uuid.5084e29a6bb24fbca5ba076eda8ecd7e
   Runtime Name: vmhba64:C0:T1:L1
   Device: uuid.5084e29a6bb24fbca5ba076eda8ecd7e
   Device Display Name: NVMe Fibre Channel Disk
(uuid.5084e29a6bb24fbca5ba076eda8ecd7e)
   Path State: active unoptimized
   Path Config: {TPG_id=0,TPG_state=ANO,RTP_id=0,health=UP}
```

已知問題

- ESXi 7.0 U3（及更新版本）、支援NVMe/FC、從ONTAP 版本號的更新版本更新至版本號的更新版本。這是因為主要NVMe中止（由ESXi 7.0 U3及更新版本發行）修正程式ONTAP 、僅從版本號支援更新到版本號

更新。請參閱相關的待處理公開報告、網址為 https://mysupport.netapp.com/site/bugs-online/product/ONTAP/BURT/1420654 以取得詳細資料。

相關連結

"TR-4597- VMware vSphere搭配ONTAP VMware"
"VMware vSphere 5.x、6.x及7.x支援NetApp MetroCluster 功能（2031038）"
"VMware vSphere 6.x與7.x支援搭配NetApp®SnapMirror®營運不中斷（SMBC）"

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